STUDENT PERFORMANCE ANALYTICS SYSTEM

Submitted by: P RAVI RAO

1. Introduction

The Student Performance Analytics System is a Python-based mini project developed to analyze students' academic performance. It processes student records stored in a CSV file and calculates total marks, averages, grades, and pass/fail status.

The system also identifies top-performing and low-performing students and performs subject-wise analysis to understand overall class performance.

2. Objective

The main objectives of this project are:

- To manage and analyze student data using Python.
- To read and process student records from a CSV file.
- To calculate total marks and average marks for each student.
- To assign grades and determine pass/fail status.
- To identify the highest- and lowest-performing students.
- To perform subject-wise and overall class analysis.
- To find the average attendance of students.
- To practice data cleaning, reusable functions, loops, NumPy, and Pandas.

3. Technologies Used

Python: Used to implement the program logic, calculations, loops, and functions.

Pandas: Used to load CSV data, create and manipulate DataFrames, clean data, and perform analysis.

NumPy: Used for numerical calculations and operations on student marks.

CSV: Used as the input data format for storing student records.

IF-ELSE STATEMENTS: used for conditional statements.

FUNCTIONS: Used for repetitive tasks.

Visual Studio Code: Used to write, execute, and test the Python program.

4. Dataset Description

The project uses a CSV file named student_data.csv containing approximately 40 student records.

The dataset contains the following columns:

- Student_ID — Unique student identifier.
- Name — Student name.
- Department — Student's department.
- ML — Machine Learning marks.
- Java — Java programming marks.
- Maths — Mathematics marks.
- OS — Operating Systems marks.
- Attendance — Student attendance percentage.

The four subjects are evaluated out of 100 marks each, giving a maximum total of 400 marks per student.

5. Implementation

5.1 Data Loading

The program reads student records from a CSV file using Pandas and stores them in a DataFrame for further processing.

5.2 Data Cleaning

The program performs data-cleaning operations to improve data consistency. These operations include removing unwanted spaces from names, standardizing department names, and handling missing or incorrectly formatted values.

Invalid attendance values are also checked and handled according to the rules implemented in the program.

5.3 Total and Average Calculation

The total marks are calculated by adding the marks obtained in all four subjects.

The student average is calculated by dividing the total marks by four.

5.4 Grade Calculation

A reusable function assigns grades based on the total marks obtained by each student. The project uses the grading thresholds defined in the Python program.

5.5 Pass/Fail Analysis

The program checks the marks obtained in all four subjects. A student passes when the marks meet the defined minimum passing requirement in every subject; otherwise, the student is marked as failed.

5.6 Performance Analysis

The program identifies the highest- and lowest-scoring students and displays the top five and bottom five students based on total marks.

5.7 Subject-Wise Analysis

The program calculates the average marks for Machine Learning, Java, Mathematics, and Operating Systems. These averages help identify the best-performing and weakest subjects.

5.8 Overall Class Analysis

The program calculates the overall class average using the average marks of all students. It also summarizes the number of students who passed and failed.

5.9 Attendance Analysis

The Program Calculates Average Attendances of Students. Student with the highest attendance and student which have very low attendance.

6. Key Features

- CSV-based student data processing.
- Data cleaning and preparation.
- Total and average marks calculation.
- Automatic grade assignment.
- Pass/fail classification.
- Highest- and lowest-performing student identification.
- Top five and bottom five student lists.
- Subject-wise average calculation.
- Overall class average calculation.
- Average Attendances of students.
- Reusable functions and loops.
- Saving processed results to a CSV file.

7. Output and Results

The program successfully processes student records and generates academic performance reports.

The output includes:

- Highest-scoring student and their total marks.
- Top five highest-scoring students.
- Bottom five lowest-scoring students.
- Pass/fail statistics.
- Average marks for each subject.
- Overall class average.
- Grades and individual student averages.
- Average Attendances of students.
- Student with highest and lowest attendance.

```
Highest Marks:
Student_ID      Name Department  ML  Java  Maths  OS  Attendance  Total  Average pass/fail Grades
      S132 Anvi Shaw         IT  97    93     96  98          79    384     96.0      PASS     A+

Top 5 Highest Marks:
Student_ID           Name Department  ML  Java  Maths  OS  Attendance  Total  Average pass/fail Grades
      S132      Anvi Shaw         IT  97    93     96  98          79    384     96.0      PASS     A+
      S112     Meera Nair        CSE  96    94     91  97          99    378     94.5      PASS     A+
      S106     Ananya Das         IT  94    91     89  95          98    369    92.25      PASS     A+
      S128 Riya Mukherjee   CSE-AIML  89    95     92  91          84    367    91.75      PASS     A+
      S118   Sanjana Bose        CSE  91    89     94  92          95    366     91.5      PASS     A+

Top 5 Lowest Marks:
Student_ID           Name Department  ML  Java  Maths  OS  Attendance  Total  Average pass/fail Grades
      S119    Manish Yadav   CSE-AIML  36    44     33  40          61    153    38.25      FAIL      F
      S105     Arjun Patel        CSE  32    48     41  35          65    156     39.0      FAIL      F
      S129   Harsh Vardhan         IT  39    42     37  45          64    163    40.75      FAIL      D
      S113      Aditya Roy   CSE-AIML  41    39     52  46          74    178     44.5      FAIL      D
      S137 Sahil Choudhary   CSE-AIML  43    47     51  39          69    180     45.0      FAIL      D
```

```
ML Average Marks:  69.6
Maths Average Marks:  71.8
OS Average Marks:  72.25
Java Average Marks:  72.1

Overall Class Average:  71.44

Subject With the Highest Average: OS : 72.25
Subject With the Lowest Average: ML : 69.6
```

```
Avg Attendance of Students: 83.18

Highest Attendance:
      Name  Attendance
Meera Nair          99

Lowest Attendance:
        Name  Attendance
Manish Yadav          61
```

```
Pass/Fail Statistics:
Total Students: 40
Total Passed: 33
Total Failed: 7
```

```
Highest Marks:
Student_ID      Name Department  ML  Java  Maths  OS  Attendance  Total  Average pass/fail Grades
      S132 Anvi Shaw         IT  97    93     96  98          79    384     96.0      PASS     A+

Top 5 Highest Marks:
Student_ID           Name Department  ML  Java  Maths  OS  Attendance  Total  Average pass/fail Grades
      S132      Anvi Shaw         IT  97    93     96  98          79    384     96.0      PASS     A+
      S112     Meera Nair        CSE  96    94     91  97          99    378     94.5      PASS     A+
      S106     Ananya Das         IT  94    91     89  95          98    369    92.25      PASS     A+
      S128 Riya Mukherjee   CSE-AIML  89    95     92  91          84    367    91.75      PASS     A+
      S118   Sanjana Bose        CSE  91    89     94  92          95    366     91.5      PASS     A+

Top 5 Lowest Marks:
Student_ID           Name Department  ML  Java  Maths  OS  Attendance  Total  Average pass/fail Grades
      S119    Manish Yadav   CSE-AIML  36    44     33  40          61    153    38.25      FAIL      F
      S105     Arjun Patel        CSE  32    48     41  35          65    156     39.0      FAIL      F
      S129   Harsh Vardhan         IT  39    42     37  45          64    163    40.75      FAIL      D
      S113      Aditya Roy   CSE-AIML  41    39     52  46          74    178     44.5      FAIL      D
      S137 Sahil Choudhary   CSE-AIML  43    47     51  39          69    180     45.0      FAIL      D

Pass/Fail Statistics:
Total Students: 40
Total Passed: 33
Total Failed: 7

ML Average Marks:  69.6
Maths Average Marks:  71.8
OS Average Marks:  72.25
Java Average Marks:  72.1

Overall Class Average:  71.44

Subject With the Highest Average: OS : 72.25
Subject With the Lowest Average: ML : 69.6

Avg Attendance of Students: 83.18

Highest Attendance:
      Name  Attendance
Meera Nair          99

Lowest Attendance:
        Name  Attendance
Manish Yadav          61
```

8. Challenges Faced

During development, the project involved handling missing values, inconsistent text formatting, and incorrectly formatted data. Calculating student totals, assigning grades through functions, and organizing the results into a readable format were also important implementation tasks.

These challenges helped improve my understanding of data cleaning, debugging, and data analysis using Python libraries.

9. Learning Outcomes

Through this project, I gained practical experience in:

- Python programming fundamentals.
- Variables, data types, loops, and functions.
- Reading and writing CSV files.

- Working with Pandas DataFrames.
- Performing numerical operations using NumPy.
- Cleaning and analyzing real-world-style data.
- Generating meaningful reports from datasets.
- Testing and debugging a Python project.

10. Conclusion

The Student Performance Analytics System demonstrates how Python, NumPy, and Pandas can be used to process and analyze academic data efficiently. It automates calculations, classifies student performance, and provides useful summaries of individual and overall class results.

This project strengthened my practical programming and data-analysis skills and provided a foundation for developing more advanced data science projects.